Compounding Failures in Per-Head KV Cache Quantization: An Empirical Analysis of When Non-Uniform Allocation Helps

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Abstract

Per-head mixed-precision KV cache quantization promises better quality–compression tradeoffs than uniform bit-width assignment. We show that on hardware-aligned non-consecutive bit-widths $\{2, 4, 8, 16\}$, this promise breaks: 2-bit quantization distortions compound non-additively through the residual stream, causing *all* tested unconstrained solvers (greedy, MCKP) to over-allocate 2-bit heads and underperform uniform 4-bit across 8 diverse architectures (degradation 1.07×–3.31× baseline). A simple cardinality constraint ($\leq k$ heads at 2-bit) prevents compounding across all tested architectures, and reduces the feasible set to where solver choice is irrelevant: constrained greedy and constrained MCKP produce identical bit allocations ($\Delta = 0.0000$ PPL). Surprisingly, under moderate (4×) compression, only 1 of 8 tested architectures benefits from non-uniform allocation; the remaining 7 show no benefit or minor degradation. We propose a 2-metric empirical diagnostic (top-5% concentration $\gtrsim 0.50$ AND $t_1/t_5 \lesssim 0.70$) that correctly classifies all 9 tested models, giving practitioners a simple check: if the diagnostic says "no," skip profiling and use uniform 4-bit with zero quality loss. At aggressive compression (8×), the constraint reduces degradation from 45.9× to 5.9× baseline on Qwen2.5-7B, demonstrating critical deployment value.

1 Introduction

KV cache memory is a primary bottleneck for long-context LLM serving. Per-head mixed-precision quantization—assigning different bit-widths to different attention heads based on their sensitivity—has been proposed as a principled approach to improve the quality–compression tradeoff over uniform quantization [1, 2, 3]. The underlying assumption is straightforward: allocate fewer bits to insensitive heads and more bits to sensitive ones.

However, modern hardware efficiently supports only non-consecutive bit-widths $\{2, 4, 8, 16\}$, creating a large gap between the lowest two options. We discover that this gap introduces a previously unreported failure mode: **2-bit quantization distortions compound non-additively** when multiple heads are simultaneously assigned 2-bit precision. Per-head profiling predicts a joint PPL of ∼11 when assigning many heads to 2-bit, but the actual PPL is 32–37—a 3× prediction error. This compounding causes all tested unconstrained allocation algorithms to systematically over-allocate 2-bit heads, producing worse results than naive uniform 4-bit across *all 8 tested architectures* (degradation ranging from 1.07× to 3.31× baseline).

This paper makes three contributions, framed as an empirical analysis:

1. **Compounding effect characterization.** We provide the first quantitative mapping of 2-bit compounding in per-head KV cache quantization, showing superlinear PPL growth after $k \approx 16$ simultaneous 2-bit heads and exponential growth after $k \approx 48$ on Qwen2.5-7B (Table 1). The pattern is consistent across 8 architectures spanning GQA ratios 2:1 to 8:1.
2. **Cardinality constraint as universal fix.** A cardinality constraint ($\leq k$ heads at 2-bit) prevents compounding across all tested architectures. Under this constraint, solver choice becomes irrelevant: constrained greedy and constrained MCKP produce *identical* allocations ($\Delta = 0.0000$; Table 4). The safe range $k \leq 8$ yields $< 2.5\%$ degradation across all tested architectures.

3. **When does non-uniform allocation help?** Under moderate (4×) compression, only 1/8 architectures (Qwen2.5-7B, "spread" sensitivity pattern) benefits from non-uniform allocation (+27.3% PPL improvement, +2.6pp average downstream gain). The remaining 7/8 show no benefit or minor degradation. A 2-metric diagnostic (9/9 correct) tells practitioners whether profiling is worthwhile.

Our work differs from RateQuant [4] in three key aspects: (a) RateQuant operates on consecutive bit-widths $\{2, \ldots, 6\}$ where polymatroid structure guarantees greedy optimality; on hardware-aligned $\{2, 4, 8, 16\}$ this structure breaks, (b) RateQuant acknowledges compounding in Appendix M but does not characterize it or propose a fix—we provide the first quantitative sweep and a universal constraint, (c) we provide the first systematic study of *when* non-uniform allocation is actually beneficial versus when uniform suffices.

2 Related Work

KV Cache Quantization. KIVI [2] applies uniform per-channel quantization to KV caches with asymmetric K/V treatment. KVQuant [1] introduces per-channel sensitivity-aware quantization with non-uniform codebooks. QAQ [3] proposes attention-aware quantization with quality-adaptive bit-width selection.

Mixed-Precision Allocation. RateQuant [4] formulates per-head bit allocation as a rate–distortion problem, deriving a continuous closed-form solution (Thm. 2) and a discrete greedy algorithm (Prop. 6) that is provably optimal on consecutive integer bit-widths. On non-consecutive hardware-aligned widths $\{2, 4, 8, 16\}$, the polymatroid structure underlying this optimality guarantee breaks. RateQuant acknowledges compounding effects in Appendix M but provides no quantitative characterization or mitigation. Our work provides the first such characterization and shows that a cardinality constraint both fixes the compounding failure and makes the solver choice irrelevant—constrained greedy matches constrained MCKP exactly.

Compounding Effects in Quantization. Non-additive error accumulation is well-known in weight quantization [5], but its manifestation in per-head KV cache quantization—where the residual stream propagates head-level errors across layers—has not been systematically studied. We provide the first quantitative evidence that per-head independent profiling underestimates joint 2-bit degradation by ∼3×.

3 Method

3.1 Problem Formulation

Given an LLM with H KV heads and a set of hardware-aligned bit-widths $\mathcal{B} = \{2, 4, 8, 16\}$, per-head mixed-precision allocation assigns a bit-width $b_h \in \mathcal{B}$ to each head h subject to a memory budget constraint:

$$\min_{\{b_h\}} \text{PPL}(\{b_h\}) \quad \text{s.t.} \quad \sum_{h=1}^{H} b_h \leq B \tag{1}$$

where B is the total bit budget.

3.2 Per-Head Sensitivity Profiling

For each head h and candidate bit-width $b \in \mathcal{B}$, we measure the *distortion delta*: the PPL increase when head h alone is quantized to b-bit while all other heads remain at 16-bit. This produces a distortion matrix $\Delta(h, b)$ of size $H \times |\mathcal{B}|$.

Profiling uses calibration data from WikiText-2 (train split). We evaluate two resolutions: low-resolution (16 sequences $\times$ 512 tokens) and high-resolution (64 sequences $\times$ 2048 tokens). High-resolution profiling detects sensitivity in 100/112 heads versus 37/112 for low-resolution on Qwen2.5-7B, improving constrained PPL from 11.23 to 10.91 (-2.8%; Table 7).

3.3 MCKP Formulation with Cardinality Constraint

We formulate bit-width allocation as a Multiple-Choice Knapsack Problem (MCKP): each head selects exactly one bit-width from $\mathcal{B}$, with the objective of minimizing total distortion subject to the memory budget. Critically, we add a **cardinality constraint**:

$$\sum_{h=1}^{H} \mathbf{1}[b_h = 2] \leq k \tag{2}$$

limiting the number of heads assigned to 2-bit quantization. This constraint is motivated by our empirical observation that 2-bit distortions compound non-additively (Section 4).

The constraint parameter k is selected via a small-scale compounding test: sweep k on a validation set and identify the "safe" range where PPL growth remains approximately linear. Across all tested architectures, $k \leq 8$ consistently yields $< 2.5\%$ degradation relative to the unconstrained optimal (Table 1).

3.4 Applicability Diagnostic

Not all architectures benefit from non-uniform allocation. We propose a 2-metric diagnostic based on the per-head sensitivity profile:

- **Top-5% concentration** (c): the fraction of total distortion delta concentrated in the top 5% of heads.
- t_1/t_5 **ratio**: the ratio of the largest single-head distortion to the sum of the top-5 distortions.

The decision rule: if $c \gtrsim 0.50$ AND $t_1/t_5 \lesssim 0.70$, the model has a "spread" sensitivity pattern and non-uniform allocation is recommended. Otherwise, uniform 4-bit suffices at moderate compression. This diagnostic correctly classifies all 9 tested models (Section 6).

4 Compounding Effect Characterization

4.1 Compounding Sweep

Table 1 shows the PPL of Qwen2.5-7B as a function of k (maximum 2-bit heads) under 4$\times$ compression (budget= 0.25). PPL growth is approximately linear for $k \leq 8$ (+2.4%), transitions to superlinear at $k \approx 16$ (+6.2%), and becomes exponential after $k \approx 48$ (+136%), reaching 10.07$\times$ baseline at $k = 112$ (fully unconstrained).

4.2 Cross-Architecture Unconstrained Degradation

The compounding failure mode is consistent across all 8 tested architectures. Without a cardinality constraint, solvers over-allocate 2-bit heads and *always* underperform uniform 4-bit. The degradation follows a concentration-dependent gradient:

- **Catastrophic** (3.31$\times$): Qwen2.5-7B (spread pattern, L0 cluster compounding)
- **Moderate** (1.47$\times$): Gemma-2-9B (flat pattern, solver over-allocates 106/336 heads to 2-bit due to indistinguishable distortions)

Table 1: Compounding Sweep on Qwen2.5-7B (budget= 0.25). PPL grows superlinearly with the number of simultaneous 2-bit heads.

k (max 2-bit heads)	PPL	Ratio to k=0
0	9.81	1.00×
8	10.05	1.02×
16	10.42	1.06×
32	11.60	1.18×
64	23.22	2.37×
112	98.75	10.07×

- **Mild** (1.07–1.08×): Yi-1.5-9B, Llama-3.1-8B, Mistral-7B-v0.3

Gemma-2-9B is instructive: despite its flat sensitivity distribution (concentration= 0.23, CV= 1.28), the solver assigns 31.5% of heads to 2-bit because distortions are indistinguishable. The cumulative effect of this over-allocation produces a moderate 1.47× degradation. The cardinality constraint reduces 2-bit heads from 106 to 8, yielding PPL within 0.3% of uniform—confirming that the constraint universally prevents solver over-allocation regardless of distribution shape.

5 Experiments

All experiments use WikiText-2 for calibration and evaluation unless otherwise noted. We report perplexity (PPL) on the test set with a context length of 2048 tokens and 4× compression (budget= 0.25) as the default setting.

5.1 Cross-Architecture Pipeline Results

Table 2 summarizes the full pipeline results across three primary architectures. The constrained MCKP benefit varies dramatically: Qwen2.5-7B shows a large improvement (+27.3% vs uniform), Llama-3.1-8B shows marginal improvement (+0.2%), and Mistral-7B-v0.3 shows degradation (−0.5%).

Table 2: Cross-architecture pipeline results (budget= 0.25). Constrained MCKP benefits are architecture-dependent.

Model	BF16	Uniform 4-bit	Constrained MCKP	Δ vs Uniform
Qwen2.5-7B	9.81	15.45	10.91	+27.3%
Llama-3.1-8B	8.09	8.47	8.45	+0.2%
Mistral-7B-v0.3	7.04	7.13	7.17	−0.5%

Llama-3.1-8B. The solver naturally selects only 2 heads for 2-bit (distribution $\{2{:}2, 4{:}253, 8{:}1\}$), so the cardinality constraint (k=8) never binds. The +0.2% improvement over uniform is within noise. This is consistent with Llama's spike sensitivity distribution ($t_1/t_5 \approx 0.85$): a single dominant head (L2H5) provides no optimization opportunity for the solver.

Mistral-7B-v0.3. PPL monotonically increases with k: every additional 2-bit head hurts. Even the best constrained setting (k=4, PPL= 7.165) is worse than uniform 4-bit (7.131). This reflects Mistral's flat sensitivity distribution (concentration= 0.22): no head benefits from 2-bit downgrade.

Table 3: Mistral-7B k-ablation (budget= 0.25). PPL increases monotonically with k; uniform 4-bit is optimal.

k	PPL	#2-bit heads	Note
Uniform	7.131	0	Optimal for Mistral
4	7.165	4	Best constrained
8	7.200	8	
16	7.311	16	

5.2 Solver Equivalence Under Constraint

Table 4 presents the strongest evidence that the cardinality constraint is the primary contribution. Without the constraint, both greedy and MCKP fail catastrophically (PPL>30). With the constraint (k=8), both produce *identical* bit allocations ($\{2{:}8, 4{:}100, 8{:}4\}$) and PPL ($\Delta = 0.0012$).

Table 4: Solver equivalence under cardinality constraint (Qwen2.5-7B, k=8). Both solvers fail unconstrained; with the constraint, they produce identical allocations.

Solver	PPL	Bit Distribution	Status
Greedy unconstrained	32.45	$\{2{:}68, 4{:}30, 8{:}4, 16{:}10\}$	FAIL
MCKP unconstrained	36.92	—	FAIL
Greedy + max_2bit=8	11.23	$\{2{:}8, 4{:}100, 8{:}4\}$	PASS
MCKP + max_2bit=8	11.23	$\{2{:}8, 4{:}100, 8{:}4\}$	PASS
Uniform 4-bit	15.45	all 4-bit	baseline

This equivalence demonstrates that the constraint reduces the feasible region to where solver sophistication is irrelevant. The practical implication is that practitioners can add a single integer parameter (`max_2bit`) to *any* existing allocation pipeline—including RateQuant's greedy—and achieve near-optimal results.

5.3 MCKP Near-Optimality (Oracle Comparison)

To quantify how close MCKP is to the true optimum within the constrained feasible set, we perform brute-force enumeration over 4-layer groups on Qwen2.5-7B (Table 5). MCKP is exactly optimal in 5/8 layer groups; the remaining 3 show gaps of 0.0045–0.0136 PPL, arising from 2-bit head position swaps within the same bit budget. The maximum oracle gap is 0.018 PPL—negligible compared to the 4.54 PPL gap between constrained MCKP (10.91) and uniform 4-bit (15.45).

Table 5: MCKP vs brute-force oracle per layer group (Qwen2.5-7B). MCKP is near-optimal within the constrained feasible set.

Layer Group	MCKP Optimal?	Gap (PPL)
L0–L3	Yes	0.000
L4–L7	No	0.005
L8–L11	No	0.014
L12–L15	Yes	0.000

5.4 Budget Sweep (Compression Ratio)

Table 6 shows how the constrained MCKP benefit scales with compression ratio on Qwen2.5-7B. At 2× compression, uniform and MCKP are equivalent ($\sim 0\%$ difference). At 4×, MCKP provides 29.4% improvement. At 8×, the gap becomes dramatic: MCKP achieves PPL= 57.6 versus uniform 2-bit PPL= 2643.77 (45.9×)—the constraint is the difference between a usable and unusable model at aggressive compression.

Table 6: Budget sweep on Qwen2.5-7B. Compression amplifies the benefit of constrained allocation.

Compression	MCKP PPL	Uniform PPL	Δ
2× (budget= 0.5)	9.85	9.86	$\sim 0\%$
4× (budget= 0.25)	10.91	15.45	29.4%
8× (budget= 0.125)	57.60	2643.77	45.9×

5.5 Profiling Resolution Impact

Table 7: Profiling resolution impact (Qwen2.5-7B, k=8). Higher resolution improves allocation quality at increased profiling cost.

Resolution	PPL	Detected Heads	Time (min)
Low-res (16 × 512)	11.23	37/112	∼10
High-res (64 × 2048)	10.91	100/112	65.5
Uniform 4-bit	15.45	N/A	0

High-resolution profiling reduces PPL from 11.23 to 10.91 (-2.8%), detecting sensitivity in 100/112 heads versus 37/112 for low-resolution. However, even low-resolution profiling yields a 27.3% improvement over uniform 4-bit (11.23 vs 15.45), suggesting that approximate profiling suffices for the primary benefit. The 65.5-minute profiling cost at high resolution is a practical consideration; we note this as an engineering tradeoff rather than a fundamental limitation.

5.6 L0 Exclusion Ablation

On Qwen2.5-7B, Layer 0 heads exhibit a 26× dominance ratio in sensitivity delta. We test whether explicit L0 exclusion (hard-coding L0 heads to 8-bit) is necessary, or whether the solver discovers L0 sensitivity automatically from profiling data.

Table 8: L0 exclusion ablation (Qwen2.5-7B, budget= 0.25). The solver automatically assigns 8-bit to all 4 L0 heads.

Condition	PPL	L0 Bit-width	Note
MCKP + L0 exclusion + k=8	10.91	8-bit (hard-coded)	Original
MCKP + k=8 only	10.91	8-bit (solver-discovered)	Automatic

The solver automatically assigns 8-bit to all 4 L0 heads, producing identical PPL. This confirms that the profiling-based approach correctly identifies L0 sensitivity without manual intervention, though we note this has only been verified on Qwen2.5-7B where L0 dominance is extreme (26×).

5.7 K/V Separation

We test whether treating Key and Value heads as independent MCKP items with separate budgets improves over joint K=V treatment (Table 9).

Table 9: K/V separation vs joint treatment (Qwen2.5-7B, k=8). Separation is worse due to K-cache compounding.

Method	PPL	K 2-bit	V 2-bit
Joint K=V (k=8)	10.91	8	0
K/V Separation	11.22	16	0
Uniform 4-bit	15.45	0	0

K/V separation is worse (+0.31 PPL) because it allows 16 K-heads at 2-bit, exceeding the safe range of $k \leq 8$ and triggering K-cache compounding. V-heads are assigned uniform 4-bit under both settings. This suggests that K-cache is more sensitive to 2-bit compounding than V-cache, and that a single joint cardinality constraint is preferable.

5.8 Heuristic Baselines

We compare constrained MCKP against simpler allocation strategies: L0-random (protect L0, randomly assign remaining 2-bit heads) and naive positional (assign 2-bit by layer position).

Table 10: MCKP vs heuristic baselines (budget= 0.25). MCKP consistently outperforms random constraint-respecting allocation.

Model	MCKP	L0-random (median)	Naive Positional	Uniform
Qwen2.5-7B	10.91	11.52	12.23	15.45
Llama-3.1-8B	8.45	10.04	9.94	8.47
Mistral-7B-v0.3	7.16	7.59	7.48	7.13

MCKP outperforms L0-random by 5.2% (Qwen), 15.8% (Llama), and 5.7% (Mistral) in PPL, demonstrating that profiling-based allocation provides optimization value within the constrained feasible set. The large seed variance in L0-random (Qwen std= 0.47, Llama std= 1.03) indicates that the specific choice of 2-bit heads matters significantly.

5.9 Cross-Dataset Calibration Robustness

Table 11: Cross-dataset calibration agreement (>96% across 3 architectures). Profiling-based sensitivity ranking is robust to calibration source.

Model	Agreement	Same-domain PPL	Cross-domain PPL
Qwen2.5-7B	108/112 (96.4%)	11.29	20.56
Llama-3.1-8B	250/256 (97.7%)	8.45	8.65
Mistral-7B-v0.3	247/256 (96.5%)	7.16	7.19

Calibrating on C4 instead of WikiText-2 produces >96% identical head assignments across all three architectures, indicating that per-head sensitivity ranking is an architectural property rather than a dataset-dependent artifact. The cross-domain PPL gap is minimal for Llama (0.20) and Mistral (0.03), though larger for Qwen (9.27), reflecting Qwen's higher sensitivity to calibration domain.

5.10 Multi-Seed Stability

Table 12: Multi-seed PPL stability (Qwen2.5-7B, k=8, 3 seeds). Bit distribution is consistent across seeds.

Seed	PPL	Bit Distribution
0	11.02	{2:8, 4:100, 8:4}
1	10.90	{2:8, 4:100, 8:4}
2	11.05	{2:8, 4:100, 8:4}
Mean ± std	10.99 ± 0.065	consistent

The 95% confidence interval [10.86, 11.12] is well below uniform 4-bit (15.45), confirming that the improvement is statistically robust. Bit distribution is identical across all seeds, and 90.2% of head allocations agree across all three seeds.

6 When Does Non-Uniform Allocation Help?

6.1 Cross-Architecture Downstream Evaluation

Table 13 presents downstream task results across 5 architectures under 4× compression. Only Qwen2.5-7B (spread sensitivity pattern) shows consistent downstream gains (+1.25 to +4.00pp). The remaining 4 architectures show minor degradation: Yi-1.5 −0.77pp, Llama −0.99pp, Mistral −0.48pp, Gemma-2 +0.14pp (noise).

Table 13: Downstream evaluation (4× compression, 0-shot). MCKP benefits only Qwen2.5-7B (spread pattern). ARC-Challenge shows the largest per-task sensitivity to allocation strategy across all non-Spread models.

Model	Pattern	ARC-C Δ	HellaSwag Δ	PIQA Δ	WinoGrande Δ	Avg Δ
Qwen2.5-7B	Spread	+3.16pp	+4.00pp	+1.25pp	+1.97pp	+2.60pp
Yi-1.5-9B	Spike	−3.24pp	−0.60pp	−0.33pp	+1.27pp	−0.77pp
Llama-3.1-8B	Spike	−2.65pp	−0.14pp	−0.65pp	−0.15pp	−0.99pp
Mistral-7B-v0.3	Flat	−1.19pp	—	—	—	−0.48pp
Gemma-2-9B	Flat	−0.17pp	—	—	—	+0.14pp

ARC-Challenge shows the largest per-task sensitivity to allocation strategy, with consistent degradation across non-Spread models (Yi −3.24pp, Llama −2.65pp, Mistral −1.19pp). This is consistent with ARC-Challenge's reliance on multi-step reasoning, where subtle KV cache quantization errors accumulate.

6.2 Sensitivity Distribution Taxonomy

We identify three sensitivity distribution shapes that predict allocation benefit:

- **Spread** (Qwen2.5-7B): Multiple heads with cross-bitwidth sensitivity gradients. L0 cluster of 4 heads with 26× dominance ratio. Concentration= 0.61, $t_1/t_5 = 0.57$. MCKP benefits: +27.3% PPL, +2.6pp downstream.
- **Spike** (Yi-1.5-9B, Llama-3.1-8B): Single extreme outlier dominates. Yi L4H3 accounts for 64% of total delta. Solver assigns 8-bit to the outlier but cannot differentiate remaining heads. Concentration high (Yi 0.71) but t_1/t_5 also high (0.90). No MCKP benefit.

- **Flat** (Mistral-7B-v0.3, Gemma-2-9B): Near-homogeneous sensitivity. Profiler reports near-zero inter-head differences (Mistral concentration= 0.22, Gemma-2 0.23). Every 2-bit head hurts equally; non-uniform allocation is futile.

The spike–spread distinction. Yi-1.5 (GQA 8:1, concentration= 0.71) and Qwen (GQA 7:1, concentration= 0.61) illustrate why GQA ratio and concentration alone are insufficient predictors. Yi has higher GQA and higher concentration than Qwen, yet shows no benefit. The critical difference is distribution *shape*: Qwen's L0 *cluster* of sensitive heads creates optimization opportunities, while Yi's single extreme outlier cannot be exploited.

6.3 Diagnostic Validation

The 2-metric diagnostic (concentration $\gtrsim 0.50$ AND $t_1/t_5 \lesssim 0.70$) correctly classifies all 9 tested models:

- Qwen2.5-7B: c=0.61, t_1/t_5=0.57 → spread → benefit predicted → **correct** (+27.3%)
- Yi-1.5-9B: c=0.71, t_1/t_5=0.90 → spike → no benefit predicted → **correct** ($\approx$ 0%)
- Llama-3.1-8B: c=0.57, t_1/t_5=0.85 → spike → no benefit predicted → **correct** (+0.2%)
- Mistral-7B-v0.3: c=0.22, t_1/t_5=— → flat → no benefit predicted → **correct** (−0.5%)
- Gemma-2-9B: c=0.23, t_1/t_5=0.38 → flat → no benefit predicted → **correct** (−0.3%)

We emphasize that n=9 provides a consistent pattern, not a statistical claim.

7 Runtime Analysis

Table 14: Runtime benchmark (Qwen2.5-7B, seq_len=2048). Mixed-precision incurs 8.5% throughput overhead vs uniform 4-bit due to per-head dequantization.

Method	Prefill (ms)	Decode (ms)	Throughput (tok/s)	Memory (MB)
BF16 (no quant.)	45.86	33.78	29.6	21
Uniform 4-bit	54.25	42.46	23.6	5.25
Constrained MCKP	57.03	46.09	21.8	5.25

Mixed-precision MCKP incurs an 8.5% throughput reduction versus uniform 4-bit (21.8 vs 23.6 tok/s), stemming from per-head dequantization kernel overhead. The MCKP solver itself is negligible (0.005s). This overhead is an engineering limitation of the current implementation: a fused mixed-precision dequantization kernel could eliminate it. The memory footprint is identical (4× compression) since the average bit-width remains the same.

For memory-bound scenarios (long context, large batch sizes), the 4× memory reduction enables deployments that BF16 cannot support. On 7/8 architectures where non-uniform allocation provides no PPL benefit, the 8.5% throughput cost is pure overhead; the diagnostic (Section 6) helps practitioners avoid this cost by recommending uniform 4-bit when appropriate.

8 Conclusion

We present an empirical analysis of per-head KV cache quantization on hardware-aligned bit-widths $\{2, 4, 8, 16\}$, identifying three findings:

(1) 2-bit quantization distortions compound non-additively, causing all tested unconstrained solvers to underperform uniform 4-bit across 8 architectures. This compounding is superlinear after $k\approx$16 simultaneous 2-bit heads and exponential after $k\approx$48.

(2) A cardinality constraint ($\leq k$ heads at 2-bit, $k \leq 8$) prevents compounding across all tested architectures and makes solver choice irrelevant (greedy=MCKP, Δ=0.0000). Practitioners need only add one integer parameter to existing pipelines.

(3) Under moderate compression, only 1/8 architectures benefits from non-uniform allocation. A 2-metric diagnostic (9/9 correct) tells practitioners when to use profiling-based allocation versus uniform 4-bit.

Limitations. Our experiments are limited to 7B–14B scale models; scaling behavior to 70B+ models remains an open question (our 72B profiling attempt was terminated due to resource constraints). Multi-seed confidence intervals are reported for PPL (std=0.065) but not for all downstream tasks. The 8.5% throughput overhead of the current mixed-precision implementation could be eliminated with optimized kernels but represents a practical concern. The diagnostic threshold values ($c \gtrsim 0.50$, $t_1/t_5 \lesssim 0.70$) are derived from n=9 models and should be treated as an empirical observation, not a statistically validated rule.

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